

2026-05-06 Alex Gates Briefing Outline

Purpose

This note directly answers Alex's May 5 guidance for the next briefing:

1. Articulate the minimal Tier 1 model.
2. State the model components and minimal assumptions.
3. Identify the unit of modeling and the type/domain of each variable.
4. Explain how model objects correspond to fitted data features.
5. Define a fitting plan.
6. Show how the model can return an interpretable maximizing point.

The goal for tomorrow is not to present the final dissertation model. The goal is to show a disciplined, minimal, fit-able model function that can be connected to the current basketball data and extended later.

1. Unit Of Modeling

Primary unit: (i, j, t) , where i is player, j is team / local opportunity pool, and t is season.

The empirical row is a player-season or player-team-season. The local competitive environment is the team-season roster P_{jt} . The advancement outcome is global: eventual NBA draft selection from the broader annual / cohort-level selection pool.

Working interpretation:

- Local opportunity is allocated inside P_{jt} .
- Global distinction is allocated across the whole eligible population.
- The model asks how local pool structure affects global advancement probability.

2. Minimal Model Components

The briefing should separate the **minimal Tier 1 model** from the richer set of empirical quantities we can use to estimate and diagnose it. Otherwise the model will look like it already has too many moving parts.

Core Tier 1 Structure

Minimal claim:

local environment + finite global distinction $\rightarrow$ nonlinear advancement probability

The smallest useful Tier 1 model has three objects:

Object	Role	Current basketball anchor
Y_i	global advancement outcome	eventual NBA draft selection
L_{ijt}	amalgamated local environment / local pool position	initially proxied by leave-self-out teammate quality
Λ_t	finite global selection capacity	annual NBA draft slots

The key choice is to treat L_{ijt} as the Tier 1 amalgamated object. It can contain both the upside and downside of the local pool:

$$L_{ijt} = \text{net local environment faced by player } i \text{ on team } j \text{ in season } t.$$

Conceptually, L_{ijt} folds together:

- developmental / signaling benefit from stronger teammates,
- local crowding or competition for opportunity,
- opportunity exposure constraints,
- relative position inside the local pool.

For tomorrow's briefing, these should not all be presented as separate minimal-model variables. They are better described as **subcomponents we may later decompose**.

First Empirical Proxy For The Amalgamated Term

First-pass empirical proxy:

$$L_{ijt} \approx Q_{ijt} = Q_{jt}^{(-i)}$$

where:

$$Q_{ijt} = \frac{1}{|P_{jt}|-1} \sum_{\ell \in P_{jt}, \ell \neq i} A_{\ell jt}$$

Current data forms:

- legacy `poolq_loo`,
- Tier 1 `congestion_quality`.

Interpretation:

- Q is not the whole theory.
- Q is the cleanest first proxy for local environment.
- The inverted-U can be presented as evidence that the net effect of local environment is beneficial at first and costly at the high end.

Diagnostic Decomposition Variables

These variables matter, but they are not the headline minimal model:

Quantity	Current data feature	Role in Tier 1
A_{ijt}	<code>perf</code> / PPM	own-performance baseline and ingredient for local-pool construction
C_{ijt}	<code>congestion_crowding</code>	diagnostic decomposition of local environment into crowding pressure
$C_{ijt}^{weighted}$	<code>congestion_crowding_weighted</code>	exploratory opportunity-weighted crowding diagnostic
O_{ijt}	<code>minutes</code>	opportunity / exposure diagnostic; possible mediator or control

Working interpretation:

- In Tier 1, C , weighted C , and O are ways to open up L , not separate claims that the minimal model has six independent mechanisms.
- For fitting, A may be included as a baseline/control because draft selection depends on own performance, but the theoretical mechanism of interest remains the local environment term.

Global Selection Capacity

Symbol: Λ_t

Meaning:

- Annual finite global selection capacity.
- Basketball example: approximately 60 NBA draft slots per year.

Outcome: $Y_i = 1$ if player i is eventually drafted, otherwise $Y_i = 0$.

3. Minimal Assumptions

A1. Local/Global Separation

Opportunity is local; advancement is global.

- Local pool: team-season roster.
- Global distinction: NBA draft selection.

This is the central structural distinction.

A2. Leave-Self-Out Environment

Ego’s local environment must exclude ego: $Q_{jt}^{(-i)}$.

This avoids mechanically including the player’s own performance in the peer-quality variable.

A3. Variables Must Have Interpretable Domains

Alex’s guidance: state which variables are normalized, bounded, Gaussian/standardized, or binary.

Current working domains:

Object	Type / Domain	Empirical form
Y_i	binary, $\{0, 1\}$	<code>Y_draft</code>
L_{ijt}	continuous, standardized or normalized	first proxied by <code>poolq_loo</code> or <code>congestion_quality</code>
A_{ijt}	continuous, standardized or raw positive rate	<code>perf</code> / PPM; baseline/control, not the core mechanism
C_{ijt}	continuous, standardized or normalized	decomposition diagnostic: <code>congestion_crowding</code> , <code>congestion_crowding_weighted</code>
O_{ijt}	nonnegative exposure; optionally normalized to $[0, 1]$	decomposition diagnostic: <code>minutes</code>
Λ_t	positive integer / fixed capacity	NBA draft slots

Immediate standardization rule:

- For exploratory fitting, estimate with standardized L and standardized diagnostic terms so coefficients are comparable and the model behaves numerically.
- If a variable is interpreted as a share or probability, normalize to $[0, 1]$.
- If a variable is interpreted as approximately Gaussian, use z-scores and check distribution shape.

A4. Minimal Mechanisms Only

Tier 1 intentionally includes only:

- heterogeneous local pools,
- finite global selection,
- an amalgamated local-environment term that can later be decomposed.

Tier 1 intentionally excludes for now:

- endogenous sorting,
- strategic behavior,
- dynamic adaptation,
- network topology,
- evaluator bias,
- full causal identification.

The goal is mechanism sufficiency: can the minimal structure reproduce the observed inverted-U?

A5. Shape Comes From Competing Margins

The central mechanism is that the net local-environment term can have competing margins: $\frac{\partial B(Q)}{\partial Q} > 0$, but at high local pool quality or high congestion, $\frac{\partial Q}{\partial Q} < 0$ or $\frac{\partial Q}{\partial C} < 0$.

So the advancement probability can rise with local environment at first and fall once the costly part of that same environment dominates.

4. Data Correspondence

Current basketball pipeline mapping:

Model object	Meaning	Current data feature
Y_i	eventual global advancement	<code>Y_draft</code>
L_{ijt}	amalgamated local environment / local pool position	initially <code>poolq_loo</code> or <code>congestion_quality</code>
A_{ijt}	own ability / performance proxy	<code>perf</code> from <code>PERF_METRIC</code> , currently PPM; baseline/control
C_{ijt}	decomposition of L : local crowding / roster pressure	<code>congestion_crowding</code>
$C_{ijt}^{weighted}$	decomposition of L : opportunity-weighted crowding	<code>congestion_crowding_weighted</code>
O_{ijt}	decomposition of L : opportunity / exposure	<code>minutes</code>
P_{jt}	local pool	team-season roster
Λ_t	global draft capacity	annual NBA draft slots

Current implementation status:

- `535_sports_tier_1.ipynb` now constructs `congestion_quality`, `congestion_crowding`, and optionally `congestion_crowding_weighted`.
- CELL 4 can reproduce the `530_sports_pipeline.ipynb` binned-bar shape and can switch between legacy `poolq_loo` and Tier 1 mechanism variables.
- The next modeling step is a cleaner CELL 5 regression / model-fitting block.

5. Candidate Model Functions

Minimal Diagnostic Shape Model

First diagnostic specification, using L as the amalgamated local-environment term:

$$\Pr(Y_i = 1) \approx \alpha + \beta_1 L_{ijt} + \beta_2 L_{ijt}^2 + \theta A_{ijt} + \eta_t + \varepsilon_{ijt}.$$

Expected sign pattern for inverted-U in L : $\beta_1 > 0$ and $\beta_2 < 0$.

Purpose:

- Not the final model.
- Useful because it returns a simple turning point.
- Keeps the minimal model focused on local environment and finite distinction, while allowing A as an own-performance baseline.

First empirical implementation:

- Set $L_{ijt} = Q_{ijt}$, using `poolq_loo` or `congestion_quality`.
- Interpret the curvature in L as the reduced-form net of benefits and costs of the local environment.

Decomposition / Guardrail Model

Only after the minimal shape is established, open L into diagnostic pieces:

$$\Pr(Y_i = 1) \approx \alpha + \beta_1 Q_{ijt} + \beta_2 Q_{ijt}^2 + \gamma D_{ijt} + \theta A_{ijt} + \eta_t + \varepsilon_{ijt}.$$

where D_{ijt} is one diagnostic decomposition term at a time:

- C_{ijt} : plain local crowding,
- $C_{ijt}^{weighted}$: opportunity-weighted crowding,
- O_{ijt} : minutes / exposure.

Purpose:

- Tests where the reduced-form curvature in L is coming from.
- Avoids presenting Q , C , O , and weighted C as four separate minimal mechanisms.
- Preserves the April 30 guidance: the model stays minimal; richer terms are interpretive decomposition.

Possible interaction variant:

$$\Pr(Y_i = 1) \approx \alpha + \beta Q_{ijt} + \gamma C_{ijt} + \kappa(Q_{ijt} \times C_{ijt}) + \theta A_{ijt} + \varepsilon_{ijt}.$$

This variant is useful if the question becomes whether quality is beneficial only when crowding is low, but it should be presented as a diagnostic extension, not the first minimal model.

Probabilistic / MLE Version

Once the functional form is fixed, use a binary response model:

$$\Pr(Y_i = 1) = g^{-1}(\alpha + \beta_1 L_{ijt} + \beta_2 L_{ijt}^2 + \theta A_{ijt})$$

where g^{-1} can be logit or probit.

Estimator:

- MLE for logit/probit.
- OLS/LPM remains the first diagnostic because it is transparent and easy to compare to binned means.

6. Fitting Plan

Step 1: Descriptive Shape

Fit no model first:

- bin Q ,
- plot mean `Y_draft` by bin,
- compare `poolq_loo`, `congestion_quality`, plain `congestion_crowding`, and weighted crowding.

Purpose:

- Confirm the phenomenon remains visible under each variable definition.
- Avoid fitting a model to an artifact of stale data or a bad binning choice.

Step 2: Diagnostic Least Squares

Fit $Y_i \sim L + L^2 + A$ using OLS / LPM, where L is first proxied by `poolq_loo` or `congestion_quality`.

Purpose:

- Estimate the sign and scale of the quadratic relationship.
- Recover a first-pass turning point without overloading the minimal model.

Step 3: Binary Response / MLE

Fit the same conceptual model using logit or probit: $Y_i \sim \text{Logit/Probit}(L, L^2, A)$.

Purpose:

- Respect binary outcome structure.
- Estimate a generalizable model function.
- Compare whether the fitted maximum is stable relative to LPM.

Step 4: Robustness / Specification Checks

Candidate checks:

- season fixed effects,
- team or conference controls if needed,
- cluster-robust standard errors by team or player where appropriate,
- compare equal-width bins vs quantile bins,
- add one diagnostic decomposition term at a time: plain crowding, weighted crowding, or minutes,
- print unclipped predictions before trusting any plotted model curve.

Step 5: Generalizable Function

Define a reusable function $\hat{p}(Y = 1 \mid L, A)$, with optional decomposition diagnostics.

Inputs:

- standardized L ,

- ability/performance proxy A ,
- optional diagnostic term D ,
- optional season controls.

Outputs:

- predicted draft probability,
- fitted coefficients,
- estimated maximizing L^* ,
- uncertainty interval for L^* if feasible.

7. Extracting The Maximum / Turning Point

Quadratic LPM Turning Point

If $\hat{p}(L) = \alpha + \beta_1 L + \beta_2 L^2 + \text{controls}$, holding controls fixed, then:

$$\frac{\partial \hat{p}}{\partial L} = \beta_1 + 2\beta_2 L$$

Set derivative equal to zero:

$$\beta_1 + 2\beta_2 L^* = 0$$

So:

$$L^* = -\frac{\beta_1}{2\beta_2}$$

Interpretation:

- L^* is the estimated local-environment level at which predicted advancement probability is maximized, conditional on controls.
- If L is proxied by Q , then L^* can be reported as the corresponding Q^* in the original peer-quality units.
- If L is standardized, convert the estimate back to the original units for interpretation.

Logit / Probit Turning Point

If $\hat{p}(L) = g^{-1}(\alpha + \beta_1 L + \beta_2 L^2 + \text{controls})$, and g^{-1} is monotonic, then the maximum in probability occurs at the same L^* as the maximum of the linear index when the only local-environment terms are L and L^2 .

If a later diagnostic model includes both L^2 and an $L \times D$ interaction, then:

$$L^*(D) = -\frac{\beta_1 + \kappa D}{2\beta_2}$$

This is useful as a later diagnostic because it can answer: "Does the maximizing local-environment level change when a specific decomposition term is high?"

8. What To Show Alex Tomorrow

Direct Answer To “What Is The Model?”

The model is a minimal local/global advancement model:

amalgamated local environment + finite global selection → nonlinear advancement probability

The unit is player-team-season. The local pool is team-season. The outcome is eventual global draft selection. In the first empirical implementation, the amalgamated local-environment term is proxied by leave-self-out teammate quality.

Direct Answer To “What Are The Minimal Assumptions?”

- Advancement is globally scarce.
- Opportunity is locally scarce.
- Teammate quality is leave-self-out.
- The Tier 1 local-environment term amalgamates benefits and costs of the local pool.
- Decomposition variables such as crowding, weighted crowding, and minutes are diagnostics, not separate minimal mechanisms.
- Variables used for fitting are standardized or normalized before fitting.
- Tier 1 excludes dynamic sorting and network structure.

Direct Answer To “How Does This Fit The Data?”

Current data already contains or now constructs:

- outcome: `Y_draft`,
- first proxy for L : `poolq_loo` / `congestion_quality`,
- own-performance baseline: `perf`,
- optional decomposition diagnostics: `congestion_crowding`, `congestion_crowding_weighted`, and `minutes`.

The next fitting block estimates $Y \sim L + L^2 + A$ first by OLS/LPM, then by logit/probit MLE. Diagnostic extensions add one decomposition term at a time.

Direct Answer To “What Do You Pull Out?”

The main extractable object is:

$$L^* = -\frac{\beta_1}{2\beta_2}$$

When L is proxied by Q , this can be reported as the Q^* value that maximizes predicted advancement probability. With later decomposition interactions:

$$L^*(D) = -\frac{\beta_1 + \kappa D}{2\beta_2}$$

This gives the estimated local-environment level that maximizes predicted advancement probability.

Open Decisions

- Whether `congestion_quality` should replace legacy `poolq_loo` as the first proxy for L .

- Whether `congestion_crowding_weighted` helps decompose L more clearly than plain `congestion_crowding`.
- Whether the first decomposition model should add C , weighted C , minutes, or an interaction.
- Whether opportunity (`minutes`) should be treated as a diagnostic, control, mediator, or separately estimated component.
- Whether the first briefing should emphasize LPM transparency or MLE correctness.

9. Immediate Deliverable After Briefing

Implement CELL 5 in `sports/535_sports_tier_1.ipynb`:

1. Prepare standardized variables Y, A, Q, C, O .
2. Fit minimal diagnostic LPM: $Y \sim L + L^2 + A$, first with $L = Q$.
3. Print coefficients, unclipped predictions, and L^* / Q^* .
4. Fit logit/probit equivalent.
5. Compare L^* across legacy `poolq_loo` and `congestion_quality`, then add plain/weighted crowding only as decomposition diagnostics.
6. Export one table and one figure only after the specification is stable.